

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

7th Semester of B. Pharm. Examination

University Theory Examination November/ December 2017

Production of Bulk Drugs and Drug Intermediates II (PH420.3)

Date: 01.12.2017, Friday

Time: 10.00 a.m. to 12.00 Noon

Maximum Marks: 50

Instructions:

1. Section wise separate answer books to be used.
2. Figures to right indicate marks.
3. Draw neat labeled sketches wherever necessary.

Section- I

Q-1 Answer the following:

- | | |
|--|----|
| A. Which is the undesirable characteristic of chloroform? Suggest the alternative solvent. | 01 |
| B. Define route scouting. | 01 |
| C. Justify: Shorter route of process development is much more efficient. | 01 |
| D. Why solid phase reaction vessel is shaken during coupling reaction? | 01 |
| E. How does photolabile anchor work for solid support synthesis? Give one example. | 01 |

Q-2 Attempt any TWO of the following:

- | | |
|---|----|
| A. What is route enabling process? Explain various criteria to be considered for scale up process. | 05 |
| B. Add a note on: Protecting groups for peptide synthesis. | 05 |
| C. Name various solvents and reagents used in process development and write their advantages and disadvantages. | 05 |

Q-3 Attempt any TWO of the following:

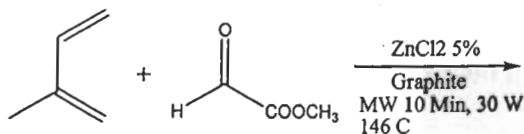
- | | |
|--|----|
| A. What is solid phase synthesis? Explain its principle and application with suitable examples. | 05 |
| B. What is the role of solid support in peptide synthesis? Describe various types of solid supports for solid phase synthesis. | 05 |
| C. Explain with suitable example: Oxidation reaction on solid support. | 05 |

Section- II

Q-4 Answer the following:

A. Complete the reaction:

01



B. What are super critical fluids? Give their advantages over usual solvents. 01

C. Write advantages of administering enantiomerically pure drugs. 01

D. What is retro synthesis? Give one example. 01

E. What are ionic liquids? Give one example. 01

Q-5 Attempt any **TWO** of the following:

A. Explain with suitable diagram, various types of microwave reactors. 05

B. Compare and contrast: conventional and microwave synthetic approaches. 05

C. Explain any two reactions performed in aqueous condition. 05

Q-6 Attempt any **TWO** of the following:

A. Add a note on microwave susceptors. 05

B. Explain with suitable examples various approaches for asymmetric synthesis. 05

C. Explain with suitable examples, asymmetric hydrogenation and oxidation reactions. 05